



# PANTOPRAZOLE 40mg INJECTION

## COMPOSITION:

Each vial contains Pantoprazole sodium equivalent to Pantoprazole 40mg.

## PRESENTATION:

A white to yellowish powder. Reconstituted solution should be clear colourless to clear yellowish solution.

## INDICATIONS:

Short term use for symptomatic improvement and healing of gastrointestinal diseases which require a reduction in acid secretion:

- Duodenal ulcer
- Gastric ulcer
- Moderate and severe reflux esophagitis
- Zollinger - Ellison - Syndrome and other pathological hypersecretory conditions

## PHARMACODYNAMICS:

Pantoprazole is a selective proton pump inhibitor. During treatment with antisecretory medicinal products, serum gastrin increases in response to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours. Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

## PHARMACOKINETICS:

Peak plasma-pantoprazole concentrations are achieved about 2 to 2.5 hours after an oral dose. The oral bioavailability is about 77% with the enteric-coated tablet formulation, and does not vary after single or multiple doses. Pantoprazole is 98% bound to plasma proteins. It is extensively metabolised in the liver, primarily by the cytochrome P450 isoenzyme CYP2C19, to desmethyl pantoprazole; small amounts are also metabolised by CYP3A4, CYP2D6, and CYP2C9. Metabolites are excreted mainly (about 80%) in the urine, with the remainder being excreted in bile. The terminal elimination half-life is about 1 hour, and is prolonged in hepatic impairment; the half-life in patients with cirrhosis was 3 to 6 hours.

## DOSAGE AND ADMINISTRATION:

The intravenous administration of Pantoprazole IV is recommended only if oral application is not appropriate. For the long term management of Zollinger-Ellison-Syndrome and other pathological hypersecretory conditions the recommended daily dose at the beginning of the treatment is 80mg pantoprazole IV. Thereafter, the dosage can be titrated up or down as needed using measurement of gastric acid secretion to guide. With doses above 80mg daily, the dose should be divided and given twice daily. A temporary increase of the dosage above 160mg Pantoprazole is possible but should not be applied longer than required for adequate acid control. With an exception of the treatment of patients with Zollinger - Ellison - Syndrome and other pathological hypersecretory conditions, the duration of treatment with the active ingredient pantoprazole should not exceed 8 weeks. In case a rapid acid control is required, a starting dose of 2x80mg Pantoprazole IV is sufficient to manage a decrease of acid output into the target range (<10mEq/h) within one hour in the majority of patients. Transition from Pantoprazole IV to the oral formulation of Pantoprazole should be performed as soon as it is clinically justified. The recommended intravenous dosage is one vial (40mg Pantoprazole) IV per day.

### Instruction of use / handling:

A ready-to-use solution is prepared by injecting 10ml of physiological sodium chloride solution into the vial containing the dry substance. This solution may be administered directly or may be administered after mixing with 100ml physiological sodium chloride solution, 5% glucose or lactated ringer's injection. Pantoprazole IV should not be prepared or mixed with solvents other than those stated. The solution should have a pH of 9. After preparation the solution must be used within 24 hours. From a microbiological point of view the product should be used immediately. If not used immediately, in-use storage times and condition prior to use are the responsibility of the user and would normally not be longer than 24 hours at not more than 25°C. The drug should be administered intravenously over 2 – 15 minutes.

### Type and duration of treatment:

As soon as oral therapy is possible, treatment with Vaxcel Pantoprazole 40mg Injection should be discontinued and the oral dosage form should be administered.

## ROUTE OF ADMINISTRATION:

For intravenous administration only.

## CONTRAINDICATIONS:

Pantoprazole IV should not be used in cases of known hypersensitivity to its constituents. Pantoprazole, like other Proton Pump Inhibitors (PPIs), should not be co-administered with atazanavir.

## SIDE EFFECTS :

See table attached.

| Frequency Organ System     | Common | Uncommon | Rare | Very Rare                    | Unknown |
|----------------------------|--------|----------|------|------------------------------|---------|
| Blood and lymphatic system |        |          |      | leukopenia; thrombocytopenia |         |

| Frequency Organ System                               | Common                                                                                 | Uncommon                                           | Rare       | Very Rare                                                                                                                        | Unknown                                   |
|------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Gastrointestinal disorders                           | Upper abdominal pain; diarrhea; constipation; flatulence; fundic gland polyps (benign) | Nausea / vomiting                                  | Dry mouth  |                                                                                                                                  | Microscopic colitis                       |
| Metabolism and nutritional disorders                 |                                                                                        |                                                    |            |                                                                                                                                  | Hypomagnesaemia<br>Vitamin B12 deficiency |
| General disorders and administration site conditions |                                                                                        |                                                    |            | peripheral edema                                                                                                                 |                                           |
| Hepatobiliary disorder                               |                                                                                        |                                                    |            | Severe hepatocellular damage leading to jaundice with or without hepatic failure                                                 |                                           |
| Immune system disorders                              |                                                                                        |                                                    |            | Anaphylactic reactions including anaphylactic shock with its typical symptoms such as dizziness, rapid pulse or profuse sweating |                                           |
| Investigations                                       |                                                                                        |                                                    |            | Increased liver enzymes (transaminases, γ-GT); elevated triglycerides; increased body temperature                                |                                           |
| Infections and infestations                          |                                                                                        |                                                    |            |                                                                                                                                  | Clostridium difficile associated diarrhea |
| musculoskeletal, connective tissue disorders         |                                                                                        | Fracture of the hip, wrist or spine.               | Arthralgia | Myalgia                                                                                                                          |                                           |
| Nervous system disorders                             | Headache                                                                               | Dizziness, disturbances in vision (blurred vision) |            |                                                                                                                                  |                                           |
| Psychiatric disorders                                |                                                                                        |                                                    |            | Mental depression                                                                                                                |                                           |
| Renal and urinary disorders                          |                                                                                        |                                                    |            | Interstitial nephritis                                                                                                           |                                           |

| Frequency Organ System                 | Common | Uncommon                                          | Rare | Very Rare                                                                                                                                                                                                                                                                                                              | Unknown                                |
|----------------------------------------|--------|---------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Skin and subcutaneous tissue disorders |        | Allergic reactions such as pruritus and skin rash |      | Urticaria; swelling of the skin or mucous membranes (angioedema); severe reactions of the skin and mucous membranes, often associated with blistering, with target lesions, (Steven-Johnson syndrome, erythema multiforme, scalded skin syndrome (Lyell's syndrome); increased sensitivity to light (photosensitivity) | Subacute cutaneous lupus erythematosus |

If you experience any side effects not mentioned in this package leaflet, please inform your doctor or pharmacist.

### Pregnancy and Lactation:

Clinical experience in pregnant women is limited. There is no information on the excretion of pantoprazole into human breast milk. Pantoprazole i.v. should only be used when the benefit to the mother is considered greater than the potential risk to the foetus / baby.

There are no known effects on the ability to drive, use machines or work without a firm support.

## SYMPTOMS AND TREATMENTS FOR OVERDOSAGE:

No symptoms of overdosage have been observed in humans. The usual rules of intoxication treatment apply in the event of the overdosage with clinical signs of intoxication.

## WARNING AND PRECAUTION:

The intravenous administration of Pantoprazole IV is recommended only if oral application is not appropriate. Pantoprazole IV is not indicated for mild gastrointestinal complaints such as nervous dyspepsia. Prior to treatment the possibility of malignancy of gastric ulcer or a malignant disease of the esophagus should be excluded as the treatment with Pantoprazole i.v. may alleviate the symptoms of malignant ulcers and can thus delay diagnosis. Diagnosis of reflux esophagitis should be confirmed by endoscopy. The daily dose of 40mg pantoprazole should not be exceeded in elderly patients or in those with impaired renal function. In patients with severe liver impairment the daily dose has to be reduced to 20mg pantoprazole. Furthermore, in these patients the liver enzymes should be monitored during Pantoprazole i.v. therapy. In case of a rise of the liver enzymes Pantoprazole IV should be discontinued.

In the presence of any alarm symptom (e. g. significant unintentional weight loss, recurrent vomiting, dysphagia, haematemesis, anaemia or melaena) and when gastric ulcer is suspected or present, malignancy should be excluded, as treatment with pantoprazole may alleviate symptoms and delay diagnosis. Further investigation is to be considered if symptoms persist despite adequate treatment. To date there has been no experience with treatment in children.

#### **Regular Surveillance:**

Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

#### **Subacute Cutaneous Lupus Erythematosus (SCLE):**

Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping Vaxcel Pantoprazole 40mg Injection. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

#### **Hypomagnesaemia:**

Severe hypomagnesaemia has been reported in patients treated with PPI like Vaxcel Pantoprazole 40mg Injection for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI. For patients expected to be on prolonged treatment or who take PPI with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

#### **Fracture:**

Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognised risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10 – 40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

#### **Clostridium Difficile Diarrhea:**

Published observational studies suggest that PPI therapy may be associated with an increased risk of *Clostridium difficile* associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

#### **Vitamin B12 Deficiency:**

Daily treatment with any acid-suppressing medications over

a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo - or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

#### **Interference with Laboratory Tests:**

Increased Chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. If the patient(s) are due to have a test on Chromogranin A level, Vaxcel Pantoprazole 40mg Injection treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (see section Pharmacodynamic). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

#### **INTERACTION WITH OTHER MEDICAMENTS:**

Changes in absorption should be observed when drugs whose absorption is pH dependent, e.g. ketoconazole, are taken concomitantly. Please note that this also applies to drugs taken a short time before Pantoprazole iv. The active ingredient of Pantoprazole iv is metabolized into the liver via the cytochrome P450 enzyme system. An interaction of Pantoprazole with other drugs or compounds which are metabolized using the same enzyme system cannot be excluded. However, no clinically significant interactions were observed in specific tests with a number of such drugs or compounds, namely carbamazepine, caffeine, diazepam, diclofenac, digoxin, ethanol, glibenclamide, metoprolol, naproxen, nifedipine, phenytoin, piroxicam, theophylline and oral contraceptive. Although no interaction during concomitant administration of phenprocoumon or warfarin has been observed in clinical pharmacokinetic studies, a few isolated cases of changes in INR have been reported during concomitant treatment in the post-marketing period. Therefore, in patients being treated with coumarin anticoagulants, monitoring of prothrombin time/ INR is recommended after initiation, termination or during irregular use of Pantoprazole. There were also no interactions with concomitantly administered antacids. If you should experience side effects, notify your doctor so that he can decide what further measure are necessary. Co-administration of atazanavir 300mg / ritonavir 100mg with omeprazole (40mg once daily) or atazanavir 400mg with lansoprazole (60mg single dose) resulted in a substantial reduction in the bioavailability of atazanavir. The absorption of atazanavir is pH dependent. Therefore Proton Pump Inhibitors (PPIs), including pantoprazole, should not be co-administered with atazanavir.

#### **STORAGE:**

Store below 30°C. Protect from light.

The reconstituted solution must be stored at below 25°C, and used within 24 hours after preparation.

**KEEP OUT OF REACH OF CHILDREN  
JAUHI DARI KANAK-KANAK**

#### **PACK QUANTITIES:**

Single vial per box and 10 vials per box.

Further information can be obtained from pharmacist, physician or the manufacturer.

Product Registration Holder / Manufacturer



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